
Simulation of Partial Melting Mantle

Chenyu Tian

CSEM, Oden Institute
The University of Texas at Austin

Todd Arbogast, Oden Institute & Mathematics, University of Texas at Austin
Marc A. Hesse, Oden Institute & Geosciences, University of Texas at Austin



Mantle melting:

1. Mantle melting is a crucial geophysical process in lithospheric evolution and plate tectonics.
2. Geophysical evidence indicates that the asthenosphere is often partially molten.
3. A thorough understanding of partial melting is fundamental to comprehending the dynamics of magmatism.

Partial Melting:

1. Modeled as a two-phase flow framework.
2. Coupled with complicated chemical interactions.
3. Compaction is essential in deciding melt extraction and segregation.

Challenge:

1. Degeneracy caused by no-melt regions poses mathematical and numerical difficulties.

Physical Model:

1. A degenerate two-phase Darcy–Stokes system describing the coupled interaction between the solid matrix and the interstitial melt.
2. A nonlinear system governing the transport of chemical species and thermal energy.
3. A eutectic phase relation characterizing the melting processes and connecting the transport and static mechanical system through enthalpy method.

Numerical Methods:

1. $H(\text{div})$ - and H^1 -conforming mixed finite element spaces for Darcy and Stokes subproblems.
2. A more versatile and flexible multilevel WENO weighting scheme to address the nonlinear transport of chemical species and energy in the presence of porous media.

1. Physical Model



Darcy-Stokes system:

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k\phi_l}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (2)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0, \quad (3)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \quad (4)$$

Deviatoric stress:

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s \left(\mathcal{D} \mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I} \right), \quad (5)$$

Transport of Chemical Species

Transport of chemical species:

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = 0, \quad (6)$$

Phase averaged quantities:

$$\overline{\rho c_i} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j}, \quad \overline{\rho c_i \mathbf{v}} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j. \quad (7)$$

Effective Velocity:

$$\overline{c_i \mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D} \right) \bar{c}_i. \quad (8)$$

Reformation:

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0 \quad (9)$$



Transport of Energy

Specific internal energy formulation:

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho Q} + \nabla \cdot \overline{K \nabla T} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v} \cdot \mathbf{g}}, \quad (10)$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity.

Temperature formulation:

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v} T}) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha \rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (11)$$

Specific enthalpy:

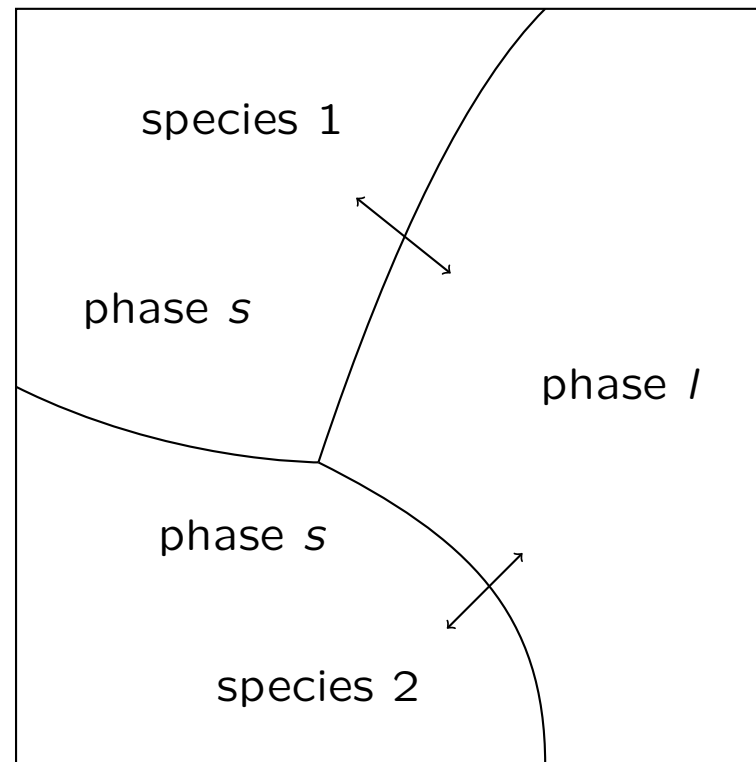
$$h_i = u_i + p_i \tilde{V}_i = u_i + P / \rho_i, \quad (12)$$

Bulk specific enthalpy:

$$\overline{H} = \overline{\rho h} / \rho c_p = \overline{h} / c_p \quad (13)$$

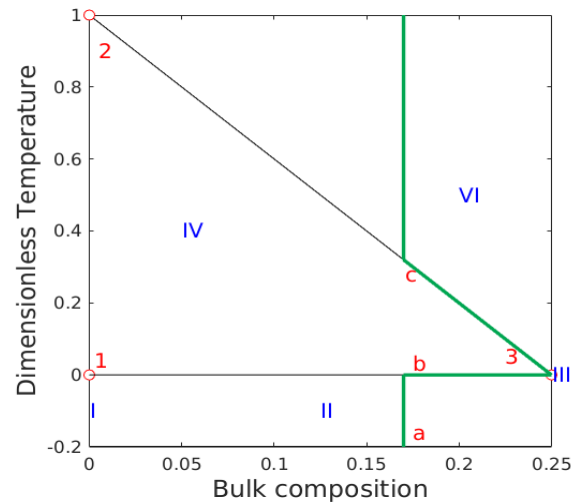


Phase transition

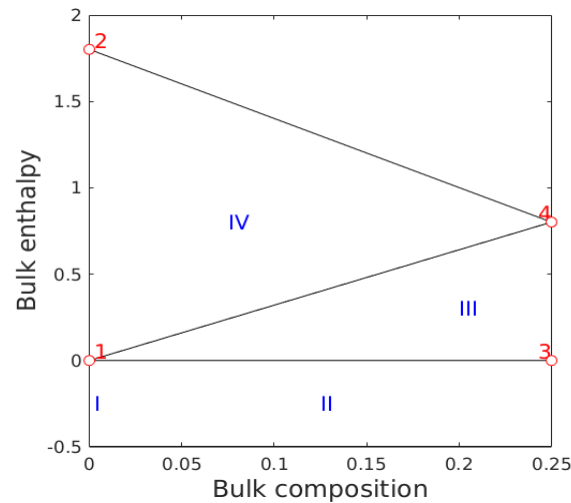


Eutectic phase behavior

Phase diagram:



Enthalpy-Composition diagram:



2. Numerical Methods



Scaled Formulation

Scaled variables:

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \quad (14)$$

Scaled Equations:

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_l} \nabla(\phi^{-1/2} \tilde{q}_l) = 0, \quad (15)$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (16)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi) \rho_r \mathbf{g}, \quad (17)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (18)$$

Well-defined condition:

$$\phi^{\Theta-1/2} \nabla \phi \in (L^\infty(\Omega))^2. \quad (19)$$



Weak formulation

Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^\phi$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla \boldsymbol{\psi}_s) - (\check{q}, \nabla \cdot \boldsymbol{\psi}_s) = -((1 - \phi)\mathbf{n}, \boldsymbol{\psi}_s) \quad \forall \boldsymbol{\psi}_s \in \mathbb{V}_S, \quad (20)$$

$$(\nabla \cdot \check{\mathbf{v}}_s, \omega) + \left(\frac{\phi^{1/2}}{1 - \phi} (\check{q}_l - \phi^{1/2} \check{q}), \omega \right) = 0 \quad \forall \omega \in \mathbb{W}_S, \quad (21)$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_r)) = 0, \quad \forall \boldsymbol{\psi}_r \in \mathbb{V}_D^\phi, \quad (22)$$

$$(\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r), \omega_l) + \left(\frac{1}{1 - \phi} (\check{q}_l - \phi^{1/2} \check{q}), \omega_l \right) = 0 \quad \forall \omega_f \in \mathbb{W}_D. \quad (23)$$

Lemma 1 (*Existence and Uniqueness*)

Assume the condition (19) holds, $0 \leq \phi \leq \phi^* < 1$. There exists a unique solution to the scaled formulation (20)–(23), and it satisfies

$$\begin{aligned} & \|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \|\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r)\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2} \\ & \leq C \{ |\rho_r| + \|\check{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \|\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{g}}_r)\|_{(L^2(\Omega))^2} + \|\check{\mathbf{g}}_s\|_{H^1(\Omega)} \}. \end{aligned} \quad (24)$$

Finite element spaces

- Reduced $H(\text{div})$ conforming space $\mathbb{V}_1^0$,

$$\begin{aligned}\mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \text{curl } \mathbb{S}_2^{\mathcal{DS}}(E), \\ \mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E);\end{aligned}\tag{25}$$

- Modified Bernardi-Raugel element,

$$\begin{aligned}\mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \\ \mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E).\end{aligned}\tag{26}$$

Homogenous spaces:

$$\begin{aligned}\mathbb{V}_{D,h,0} &= \{\mathbf{v} \in \mathbb{V}_{D,h} : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\} \\ \mathbb{V}_{S,h,0} &= \{\mathbf{v} \in \mathbb{V}_{S,h} : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\}\end{aligned}\tag{27}$$

MFEM formulation

Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h,0} + \hat{\mathbf{g}}_r$, $\check{q}_{r,h} \in \mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h} \in \mathbb{V}_{S,h,0} + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_{S,h}$ such that:

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1 - \phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}), \quad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,0}, \quad (28)$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \left(\frac{\phi^{1/2}}{1 - \phi} (\check{q}_{l,h} - \phi^{1/2} \check{q}_h), \omega_h \right) = 0, \quad \forall \omega_h \in \mathbb{W}_{S,h}, \quad (29)$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \quad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \quad (30)$$

$$(\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \left(\frac{1}{1 - \phi} (\check{q}_{l,h} - \phi^{1/2} \check{q}_h), \omega_{l,h} \right) = 0, \quad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \quad (31)$$

Local mass conservation

Element-averaged variable:

$$\phi_E = \frac{1}{|E|} \int_E \phi(\mathbf{x}) d\mathbf{x}, \quad (32)$$

Local Mass Conserved Modification:

$$(\boldsymbol{\tau}(\tilde{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\tilde{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1 - \phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \quad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,0}, \quad (33)$$

$$(\nabla \cdot \tilde{\mathbf{v}}_{s,h}, \omega_h) - \left(\frac{\phi \phi_E^{-1/2}}{1 - \phi} (\tilde{q}_{l,h} - \phi_E^{1/2} \tilde{q}_h), \omega_h \right) = 0, \quad \forall \omega_h \in \mathbb{W}_{S,h}, \quad (34)$$

$$(\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\tilde{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \quad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \quad (35)$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_{r,h}), \omega_{l,h}) + \left(\frac{\phi \phi_E^{-1}}{1 - \phi} (\tilde{q}_{l,h} - \phi_E^{1/2} \tilde{q}_h), \omega_{l,h} \right) = 0, \quad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \quad (36)$$

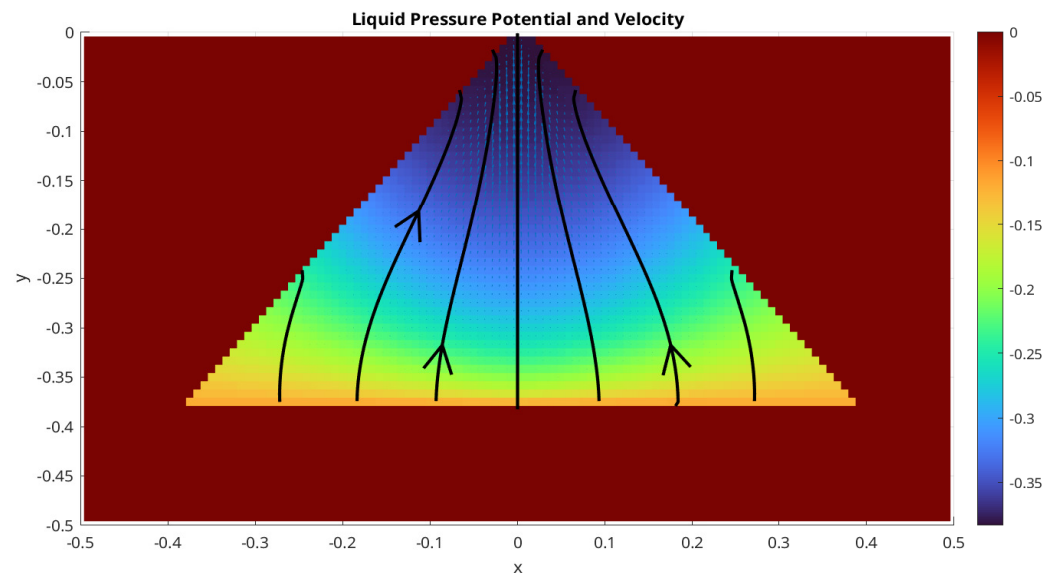
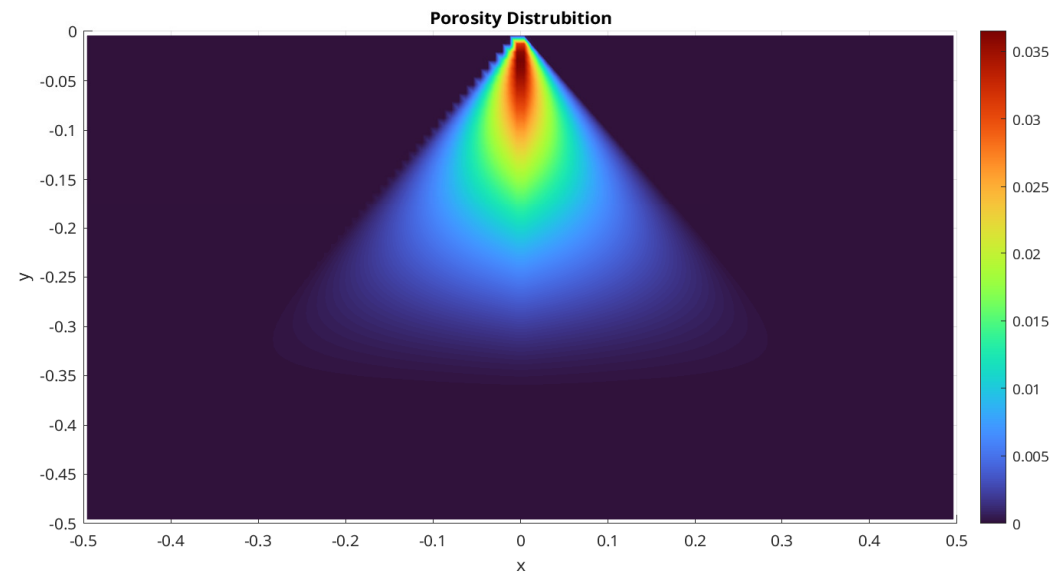


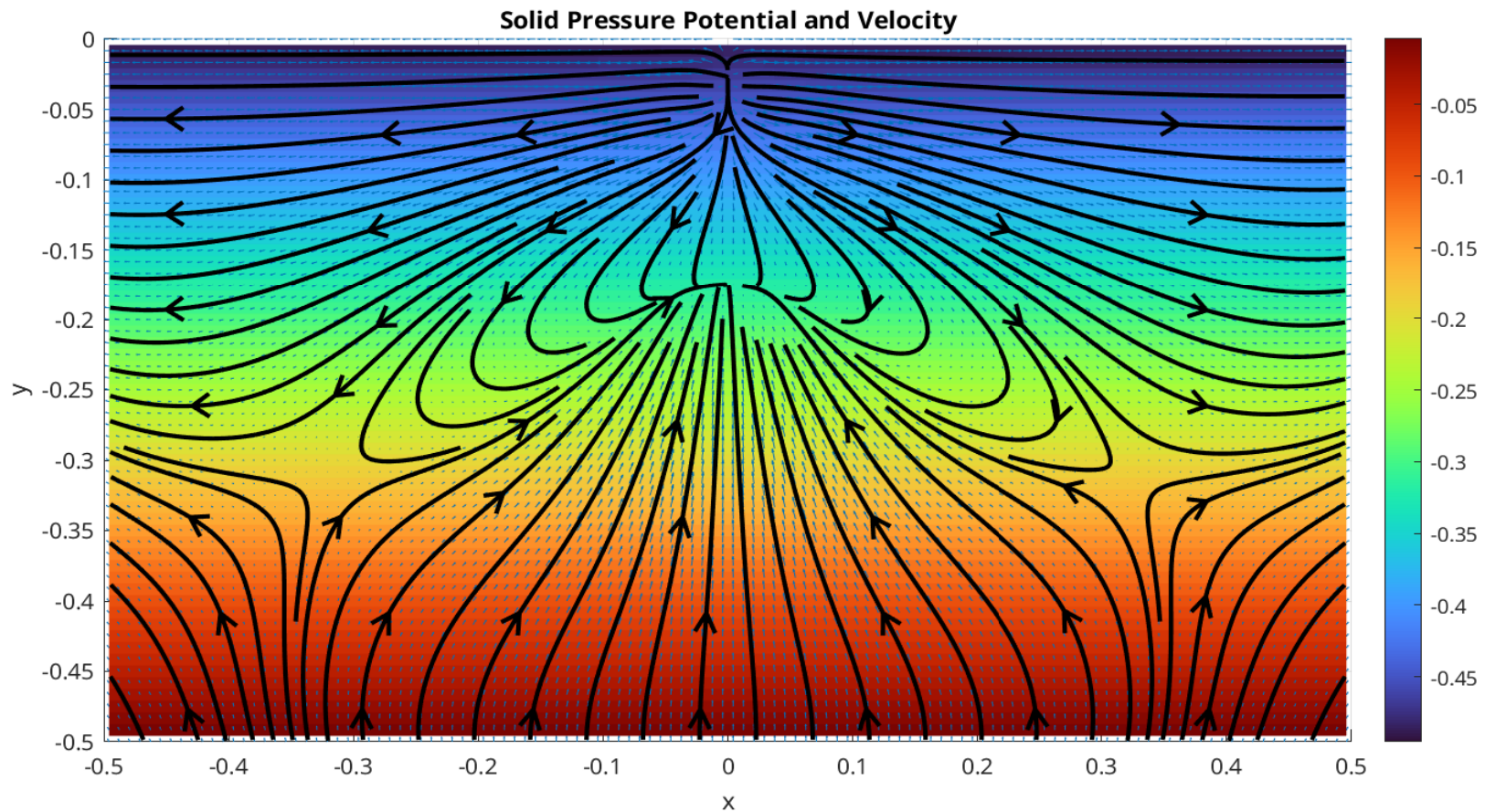
Saddle point system

Coupled Darcy-Stokes saddle point system:

$$\begin{bmatrix} \mathbf{A}_S & -\mathbf{B}_S & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_S^T & \mathbf{C}_S & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_D & -\mathbf{B}_D \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_D^T & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{y}_S \\ \mathbf{x}_D \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{G}_S \\ \mathbf{F}_D \\ \mathbf{G}_D \end{bmatrix}, \quad (37)$$

Mid-ocean ridge case





General hyperbolic conservation law:

$$\begin{cases} \frac{\partial u(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{f}(u; \mathbf{x}, t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x}, 0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \quad (38)$$

Semi-discretized form:

$$\frac{\partial \bar{u}_E}{\partial t} + \frac{1}{|E|} \int_{\partial E} \mathbf{F}(\mathbf{x}, t) \cdot \boldsymbol{\nu} d\mathbf{x} = 0, \quad (39)$$

Semi-discretized form with reconstructed value:

$$\frac{\partial \bar{u}_E}{\partial t} + \sum_{i=0}^4 \frac{|e_i|}{|E|} \sum_{g=0}^G \omega_{i,g} \mathbf{F}(\mathcal{R}_{i,g}^+, \mathcal{R}_{i,g}^-) \cdot \boldsymbol{\nu}_i = 0, \quad (40)$$

Scaled stencil polynomials:

$$Q_{s,t}^{\mathcal{S}}(x, y) = \sum_{p < s, q < t} c_{p,q} \left(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}} \right)^p \left(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}} \right)^q. \quad (41)$$

Basis polynomials:

$$\frac{1}{|E_{i',j'}|} \iint_{E_{i',j'}} \phi_{s,t}^{i,j}(x, y) dx dy = \begin{cases} 1 & (i', j') = (i, j) \\ 0 & (i', j') \neq (i, j) \end{cases}, \quad \forall E_{i',j'} \in \mathcal{S}. \quad (42)$$

Rewrite with basis polynomials:

$$\hat{Q}_{s,t}^{\mathcal{S}}(\xi, \eta) = \sum_{i,j} \bar{u}_k \hat{\phi}_{s,t}^{i,j}(\xi, \eta). \quad (43)$$

Jiang-Shu smoothness indicator:

$$\begin{aligned}
 \sigma_{JS} &= \sum_{1 \leq |\alpha| \leq r} \frac{h_S^{2|\alpha|}}{|E_0|} \iint_{E_0} (\mathcal{D}^\alpha Q(x, y))^2 dx dy \\
 &= \frac{h_S^2}{|E_0|} \sum_{1 \leq |\alpha| \leq r} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi, \eta))^2 d\xi d\eta \\
 &= \frac{h_S^2}{|E_0|} \sum_{1 \leq |\alpha| \leq r} \iint_{\hat{E}_0} \left(\sum_{ij} \bar{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi, \eta) \right)^2 d\xi d\eta \quad (44) \\
 &= \frac{h_S^2}{|E_0|} \sum_{ij} \sum_{i'j'} \bar{u}_{ij} \bar{u}_{i'j'} \sum_{1 \leq |\alpha| \leq r} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi, \eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi, \eta) d\xi d\eta \\
 &= \frac{h_S^2}{|E_0|} \sum_{ij} \sum_{i'j'} \bar{u}_{ij} \bar{u}_{i'j'} \sigma_{ij}^{i'j'},
 \end{aligned}$$

- *Simple smoothness indicator*

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\bar{u}_k - \bar{u}_0)^2, \quad (45)$$

where $\bar{u}_0$ is the cell averaged solution in the target cell. This simple smoothness indicator is designed to reduce the computational cost and to deal with distorted stencils on unstructured meshes.

However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator.

- *Polynomial smoothness indicator*

$$\sigma_P = \sum_{\alpha} \zeta_{\alpha, \alpha} c_{\alpha}^2 \quad (46)$$

which is obtained by dropping all the cross terms appeared in the formulation (??). This alternative serves as a good approximation to the classic Jiang-Shu smoothness indicator while the computing efficiency is greatly improved.

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q^{S_l}(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \quad (47)$$

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \quad (48)$$

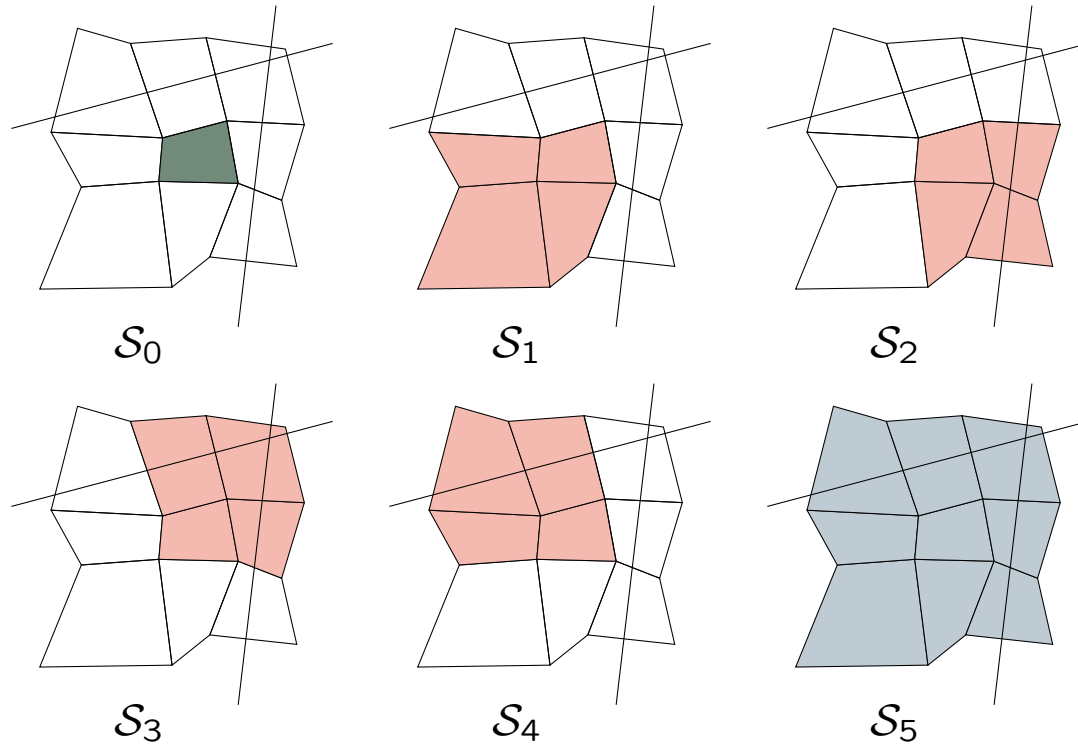
Biasing factor:

$$n(r_l) = n_l = \begin{cases} 1, & \text{if } r_l = 1, \\ 3, & \text{if } r_l = 2, \\ 4, & \text{if } r_l \geq 3. \end{cases} \quad (49)$$

Renormalized scaling weights:

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \quad (50)$$

Identifying shock



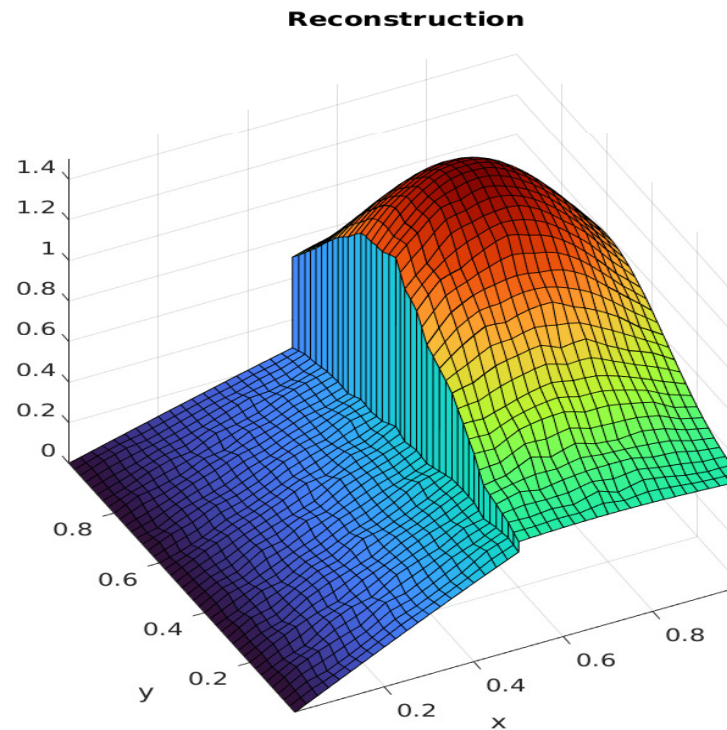
A multi-stage explicit SSP Runge-Kutta method:

$$\begin{aligned} u^0 &= u^n, \\ u^{(i)} &= \sum_{j=0}^{i-1} \alpha_{i,j} u^{(j)} + \Delta t \beta_{i,j} \tilde{\mathbf{F}}(u^{(j)}), \quad 1 \leq i \leq m \\ u^{n+1} &= u^m, \end{aligned} \tag{51}$$

where $\sum_j^{i-1} \alpha_{i,j} = 1$.

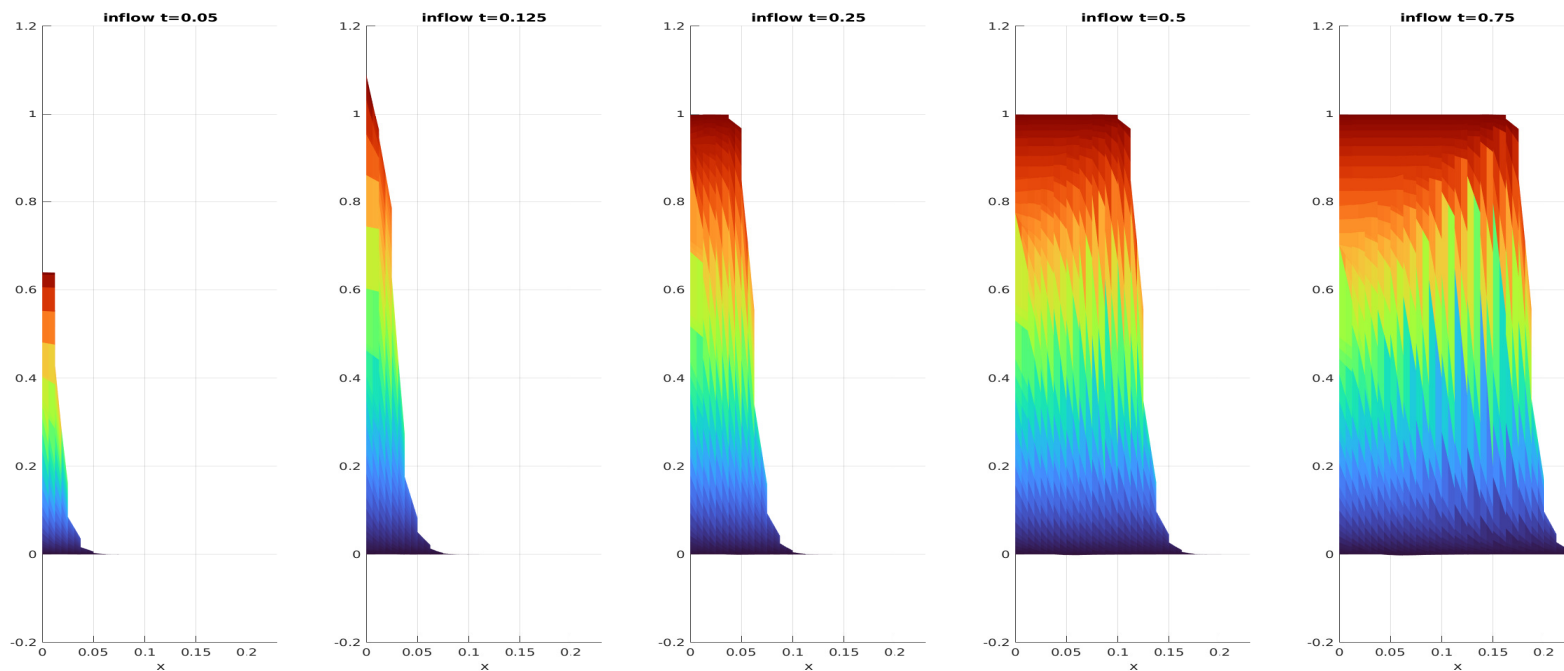
Reconstruction example

$$u(x, y) = \begin{cases} \sin(x) \cos(y) & x \leq 0.5, \\ \sin\left(\frac{(x + 0.3)\pi}{2}\right)^6 \cos\left(\frac{(y - 0.6)\pi}{2}\right)^6 + 0.5 & x > 0.5. \end{cases} \quad (52)$$



Inflow boundary stabilization

$$u_D(0, y, t) = \begin{cases} \sin^2(2\pi y), & y \leq 0.25, y \geq 0.75, \\ 1, & 0.25 < y < 0.75. \end{cases} \quad (53)$$



Rarefaction example

$$u(x, y, t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x - 0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \quad (54)$$

